

Date	Topics	Reading	Project
1/20-1/23	CLASS INTRO	Web	Software Installation
1/27-1/30	Beginning Unix	LCL Ch. 1-4, 12	Unix Lab 1
2/3-2/6	Advanced Unix I	LCL Ch. 5-11, 16, 18	Unix Lab 2
2/10-2/13	Advanced Unix II	LCL Ch. 19, 24-30	Unix Lab 3 (BASH)
2/17-2/20	Basic R Stats	R Cook	Unix Project (DUE FRIDAY @ MIDNIGHT)
2/24-2/27	R for Bioinformatics	R Bio	
3/3-3/6	Advanced Python I	APB	R Project (DUE FRIDAY @ MIDNIGHT)
3/10-3/13	Advanced Python II	APB	
3/17-3/20	Advanced Python III	APB	<i>OOP Wizard HOMEWORK (DUE BEFOR LECTURE ON TUES)</i>
3/24-3/27	Jupyter & Python Libraries (Pandas, Biopython, Scikit-learn)	ARTICLE ON JUPYTER NOTEBOOKS: https://www.theatlantic.com/science/archive/2018/04/the-scientific-paper-is-obsolete/556676/	Python part 1: OOP Project (DUE FRIDAY @ MIDNIGHT)
3/31-4/3	SPRING RECESS		
4/7-4/10	Marker Gene Microbial Communities (16S, ITS)	Readings*	QIIME 2 Python part 2: Adv. Libraries Proj (DUE FRIDAY @ MIDNIGHT)
4/14-4/17	RNA Seq Analysis	Readings*	RNA Seq Lab;
4/21-4/24	Shotgun Metagenomics & Assembly	Readings*	Metagenomics Lab Exercises
4/28-5/1	AI: Machine Learning	Readings*	Random Forests, Neural Networks
5-May	WORK ON PROJECTS (TBD)		Bioinfo Project & Exercises (DUE FRIDAY @ MIDNIGHT)
7-May	FINAL EXAM DAY		Final Project (DUE @ MIDNIGHT)

* Readings = Classic Papers on RNA Seq; 16S; Metagenomics and AI
MetagenomeClassicPapers.xlsx

Lecture(s)	Notes
UNIXlect1.1.ppt UNIXlect1.2	UNIXlect1.2notes
UNIXlect1.2 Continued	unix-file-permission.pdf
UNIXlect3.1BashScripting.pptx	Unixlect3_Old.pdf; UNIXlectgrep_

R Studio and R notebooks

Lectures: OOP1 Tues, OOP2 Thurs,
Inclass Thurs character_class.py



Python: Project involving pandas, Biopython and other libraries

MarkerGene16SAmpliconLect1	Exercises: SangerSeqExercise
MarkerGene16SAmpliconLect2	
Bob Zeller Lectures	Bob produced Zoom lectures!
ShotgunMetagenomicsLect1	Exercises:
ShotgunMetagenomicsLect2	DeBruijnGraphExercise_SOLO
Bioinfo Project: Choose 2	Choices: (1) RNA Seq (2)
Bioinformatics topics	QIIME2 (3) Metagenomics
	Pipeline (4) Machine Learning
	Analysis
Date of final exam.	

Lab(s)

<http://www.ee.surrey.ac.uk/Teaching/Unix/>

Tutorials 1-4

<http://www.doc.ic.ac.uk/~wjk/>

examples.png

Lab: person.py and wizard_class_HW.py for Tues.

Coding: CleanDenoiseASVExercise; QIIME2 Tutorial
Moving pictures (or Parkinson's)

R Seurat tutorial and 2700 PBMC vignette
MetaExercise.txt, MetagenomicsKmerProj

MetagenomicsLab Dropbox Link

<https://satijalab.org/seurat/>

MetagenomicsLab Dropbox Link

MetagenomicsLab Dropbox Link

Lecture 1 and 2 & 6 Unix Labs 1.1, 1.2 and UnixLab1.3_Vi_Emacs.docx